

RAG ASSIGNMENT

You are an engineer tasked with creating an AI agent that reads engine manuals in PDF format and provides accurate maintenance instructions. This companion will assist technicians in troubleshooting issues and performing regular maintenance on engines widely used in the agriculture industry, particularly for powering heavy farm machinery such as tractors, combine harvesters, and irrigation systems.

Recommendations:

- **Exact Responses:** The AI agent must output answers that match the wording in the manuals exactly, without paraphrasing or interpretation.
- **Manual-Specific Answers:** Each manual focuses on a specific engine. When a question applies to both manuals, the system must first ask a clarifying question (e.g., "Do you mean the Caterpillar 3500 engine or the Waukesha VGF engine?") before providing the detailed answer.
- **Complete Responses:** Answers should include all steps and details as provided in the manual, without omitting any information.
- **Response format:**
 - **Answer:** The content exactly as it appears in the manual.
 - **PDF Page Number:** The corresponding PDF page number (not the manual's internal numbering).

Deliverables:

Part 1

Create a flowchart that illustrates:

1. The architecture of the system.
2. Potential issues, challenges or limitations.

Part 2

Create a Prototype (A quick/ rough draft is sufficient for now as the goal is to identify the right references in the manuals):

You will receive two PDFs (part of a larger library) containing information about engines. Using this information, the system must answer the following question correctly:

"How to clean and maintain the air filter?"

Tip: Relevant info is included in both manuals.

Where the relevant info appears in the Manuals:

<https://storage.googleapis.com/rag-agents-case-study/case-study-1/Caterpillar-3500-generator-sets-operation-and-maintenance-manual.pdf> - page 122-125

https://storage.googleapis.com/rag-agents-case-study/case-study-1/Waukesha%20VGF_f18g.pdf - page 248-250